Software Requirements Specification

Field	Your entry
Team name	Pufai&friends
Project title	KlaiBaan — a space for sharing feelings among people who are far from home
SDLC chosen + reason	Waterfall — chosen because the team consists of a single developer (a first-year student) without prior experience managing changing requirements mid-project. The MVP scope is small and clearly defined (posting and commenting), which suits fixing requirements early and proceeding phase by phase (analysis → design → implementation → testing → deployment) with minimal iteration
Date / version	v1.1 [11 November] v1.0 was submitted at M2
GitHub repo	https://github.com/6931503041-cmd/Klaibaan-srs

1. Introduction

1.1 Purpose and scope

People who study or work far from home, whether domestic students who have moved to a new city, exchange students abroad, or professionals who have relocated — commonly experience homesickness, yet often lack an appropriate outlet for these feelings. Friends in their new environment may not fully understand the context, and they may not want to worry family back home. As a result, this sense of isolation is often kept to oneself.

KlaiBaan is a community website that allows users to create posts describing their feelings or the challenges they face while living away from home, and allows others to comment with encouragement or share similar experiences. The goal is not to provide professional counseling, but to help users feel that **they are not alone in what they are experiencing**.

The MVP consists of five core features: create/view posts, comment on posts, a tagging/filtering system by feeling category, anonymous posting, and a shared-feeling reaction with reporting of inappropriate content. Sign-up and log-in support these features, and two small engagement features (a daily encouragement card and a daily question). Success is measured as follows: a user who creates a post receives at least one comment within 24 hours.

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
User seeking to share feelings (poster)	primary user	Post their feelings without being judged and feel that someone is listening.	Being mocked or receiving hurtful comments; their identity being exposed without consent.

User offering encouragement	primary user	Easily find posts they can relate to (e.g., a similar situation)	Wasting time scrolling to find relevant posts
Moderator	secondary stakeholder	Tools to remove or hide inappropriate content in a timely manner	Harmful content (e.g., self-harm indicators) slipping through unnoticed
Mental health support organization / hotline	institution or supplier	The website does not misrepresent itself as a substitute for professional counseling services	Users mistakenly believing the website is an emergency mental health service

2. Overall description

IN SCOPE (MVP this term)	OUT OF SCOPE (explicit promise)
• FR-1 Create/edit a post • FR-2 Comment on a post • FR-3 Tag/filter posts by category • FR-4 Anonymous posting • FR-5 Shared-feeling reaction button + report inappropriate posts • Supporting (not FRs) : sign-up,log-in,log-out, daily encouragement (pre-written messages), daily question	• Private messaging between users • Professional counseling , or an AI chatbot giving psychological advice • Friend/buddy-matching system • Real-time video calls or community event • Payments or premium subscription features • Mobile app (web only

2.1 Product context — what it does / does not

2.2 Assumptions and constraints

- Assumption: every user has an email address or account used for sign-up to prevent spam, even though posts may be displayed anonymously to other users.
- Constraint: solo development team (a first-year student) — features must remain simple and must not depend on complex infrastructure (e.g., real-time chat, advanced AI moderation) in this initial phase.
- Constraint: personal data stored — limited to email/display name and the content of posts and comments; no other identifying information (e.g., national ID number) is stored.
- Constraint: 10-week development window shared with four other milestones (M1–M5); under the Waterfall SDLC, scope must therefore be fixed from M2 onward.
- Constraint: a minimum moderation mechanism (report + hide post) must be included in the MVP, given that the content concerns feelings and mental well-being.
- Constraint (prototype): there is no backend. Accounts,posts and comments are stored in the browser (localStorage).

3. Functional requirements

Format: As a <role>, I want <goal>, so that <benefit>.

FR-1 — Create / edit a post

***User story**: As a user experiencing homesickness.* ***I want** to create a post describing my feelings or the challenges I'm facing, along with selecting a category, **so that** I can express myself and give others who relate a chance to respond.*

***Pain this traces to**: Users lack an appropriate space to express feelings of homesickness, as they fear those around them will not understand, or that their family back home will worry.*

Acceptance criteria:

•Given a title, content, and category have been entered, when the user submits the post, then the post immediately appears in the feed along with the posting timestamp.

•Given an existing post owned by the user, when the user edits the content, then the updated content is displayed immediately with an "edited" label, and no duplicate entry is created.

•Fail case : given the post text is empty or no category is selected, when the user taps post, then no post is created.

FR-2 — Comment on a post

***User story:** As a user reading another person's post.**I want to** comment with encouragement or share a similar experience, **so that** the post's author feels heard and knows they are not alone.*

***Pain this traces to**: People experiencing homesickness often feel they are the only ones who feel this way, even though many others have gone through similar experiences.*

Acceptance criteria:

•Given a post open for comments, when the user types a comment and submits it, then the comment immediately appears beneath the post along with the commenter's display name.

•Given a comment that has already been submitted, when the comment's author deletes their own comment, then the comment is immediately removed from the post.

•Fail case : given an empty comment, when the user submits it, then no comment is added.

FR-3 — Tag / filter posts by category

***User story**: As a user looking for posts I can relate to**, I want to** filter posts by category (e.g., "missing home food," "difficulty adjusting," "missing family"), **so that** I can more quickly find posts that match my own situation.*

***Pain this traces to**: As the number of posts grows, users must scroll through all posts to find ones relevant to their situation, wasting time and potentially missing relevant content.*

***Acceptance criteria**:*

•Given a feed containing posts across multiple categories, when the user selects a single tag, then the feed displays only posts containing that tag.

•Given no posts exist under the selected category, when the user applies that filter, then the system displays a clear empty-state message rather than a blank screen.

FR-4 — Anonymous posting

***User story :** As a user who does not wish to reveal my identity, **I want to** choose to post without displaying my real name or account, **so that** I feel safe sharing deeper, more vulnerable feelings without worrying that acquaintances will see them.*

***Pain this traces to:** Many users hesitate to post certain deeply personal topics because they do not want people they know to identify them as the author, even though they want to express these feelings.*

Acceptance criteria:

•Given the "post anonymously" option is enabled when creating a post, when the post is published, then the displayed name on the post is a generic alias (e.g., "Anonymous User") rather than the user's real account name.

•Given an anonymous post, when another user views it, then there is no way to trace it back to the poster's real account through the UI.

•Fail case : given an anonymous post, when another user opens it or its comments, then the author's name,email and ID are not shown anywhere.

FR-5 — Shared-feeling reaction button and reporting inappropriate posts

***User story:** As a user reading a post, **I want to** click an "I understand" button to express a shared feeling without needing to write a comment, and to be able to report inappropriate posts, **so that** the post's author feels acknowledged even without comments, and the community stays safe from harmful content.*

***Pain this traces to:** Not everyone is comfortable writing a comment, but may still want to convey "I understand." At the same time, a platform centered on vulnerable emotions carries a risk of hurtful comments or dangerous content that requires a way to be managed.*

Acceptance criteria:

•Given any post, when the user clicks the "I understand" button, then the reaction count increases by one and the button's state changes to indicate it has already been clicked.

•Given a post or comment that has met the defined reporting threshold (or has been confirmed by a moderator), when the system processes the report, then the content is immediately hidden from the public feed pending moderator review.

•Fail case : given a post or comment that has been reported 3 times, when the 3 rd report is submitted, then it is hidden from the public feed pending moderator review.

•Fail case : given a user views their own post or a post they already reported, when they tap report, then no report is counted and a message is shown.

4. Non-functional requirements

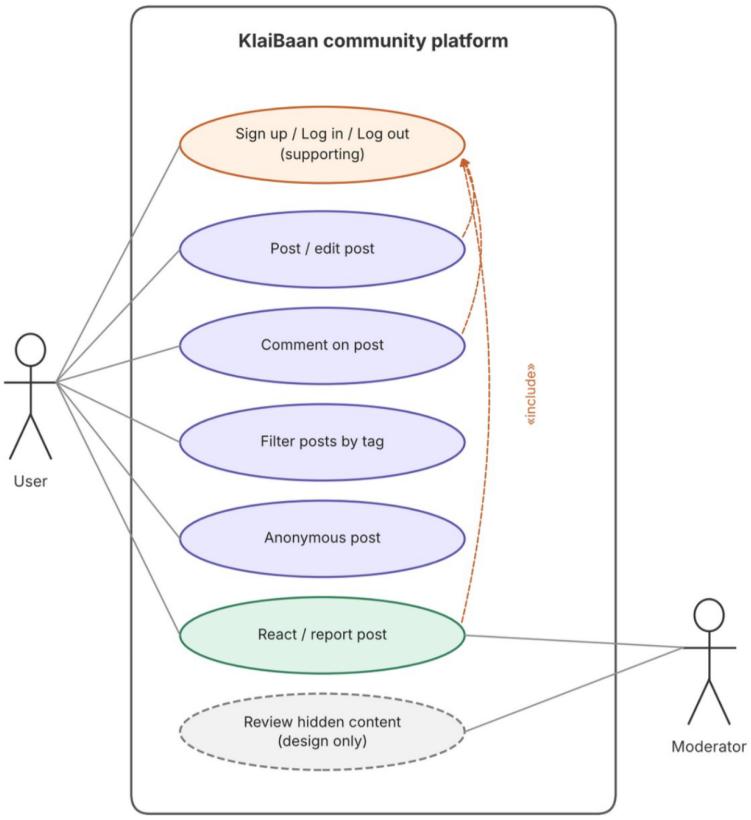
ID	Kind	Statement (testable)	How we will check
NFR-1	performance	The post feed page must be loaded within 2 seconds on the general internet.	Set the timer with the mobile clock in the classroom 3 times.
NFR-2	performance	Sent comments must appear under the post within 1 second after pressing send.	Compare the time when you press the button with the time when the UI displays the comment.
NFR-3	usability	First-time users must successfully create their first post within 1 minute without asking anyone.	Experiment with timers with users who have never seen the website before.
NFR-4	reliability	The system must keep all posts/comments even if the user refreshes the page or closes the browser and reopens it.	Close the browser and open it again. Check that all posts/comments are still there.
NFR-5	security / privacy (required)	Users who post anonymously must not have real account information leaked through any API response.	Check the network request/response of the anonymous post that there is no real user ID.
NFR-6	security	Passwords must be at least 8 characters and stored only as a hash. Log-in is locked for 30 seconds after 5 wrong attempts in a row.	Enter a wrong password 5 times and check the lock message. Open Local Storage and check that no plain password is saved.

5. Use cases

5.1 Use-case list

Use case	Actor	Related FR	Brief success story
UC-1 Post / edit post	user	FR-1	The user typed the post "At home atmosphere" with the tag "Rural" and successfully posted.
UC-2 Comment on post	user	FR-2	Another user commented "I understand, I am" under that post.
UC-3Filter posts by tag	user	FR-3	Users select the tag "Inadjustable" and see only related posts.
UC-4 Anonymous post	user	FR-4	Users turn on incognito mode and post deep stories without worry.
UC-5 React / report post	user	FR-5	A reader taps " I understand" and the count goes up. Another reader reports a hurtful post and picks a reason. After 3 reports the system hides the post to wait for the moderator.
UC-6 Sign up / log in / log out	user	Supporting	A new user signs up with an email and password, logs out,and logs in again. Log-in is needed to post, comment and report.
UC-7 Review hidden content	Moderator	FR-5	The moderator opens the list of hidden posts and comments and restores or removes each one. This is designed in KlaiBaan_Report_Screens.html and is not connected to the main prototype.

5.2 Use-case diagram



6. Out of scope

1.No payment: The website does not have any system to pay or spend money.

2. No private chat: Users cannot send private messages or talk to each other directly in a private chat. They can only comment under the posts.

3. Web only, no mobile app: This project is only a website. There is no mobile application for iOS or Android smartphones.

4. No mental health guarantee: This website is only a place for users to talk and share feelings together. The developers are not responsible for any mental health problems of the users.

5. No buddy matching : The site does not match users with each other.

6. No AI advice : the daily encouragement card shows pre-Written messages with a note that it is not medical advice. No AI model gives psychological advice.

7. Traceability matrix (Golden Thread)

Only the requirement most central to the platform's core purpose is filled in completely (following the StudyMate example, where FR-3 was fully traced); the remaining rows are left open until their corresponding week is reached.

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
Users lack an outlet to express homesickness	FR-1 Create a post	UC-1. Post entity : text, category, time, edited flag.	New-post screen (text,category, anonymous switch); edit post; "edited" label	AC1 to AC3. Test T10.
Users feel they are the only ones who feel this way	FR-2 Comment on a post	UC-2.Comment entity linked to its post; replies linked to a comment.	Comment box under a post;replies;delete own comment	AC1 to AC3. Test T11.
Difficulty finding posts relevant to one's own context	FR-3 Tag/filter	UC3-Category list; each post has a category.	Category chips on the home screen; empty-state message	AC1 to AC2. Test T12.
Fear of being recognized when posting deeply personal content	FR-4 Anonymous posting	UC-4 Anonymous flag on the post; alias Shown instead of the name.	"Post anonymously" Switch; alias in the feed	AC1 to AC3. Test T13.NFR-5.
No way to manage harmful content; people cannot say "i understand" without writing	FR-5 Reaction + report	UC-5,UC-7. Reaction count; report with a reason; hide at 3 reports; moderator review.	"I understand" button; report sheet; hidden-post state; moderator queue	AC1 to AC4. Test T14,T14b.
Supporting: spam and ownership of content (assumption in 2.2)	Accounts (not an FR)	UC-6. Users stored in the browser: email, display name, hash + salt.	Sign-up,log-in and log-out screens; log-in needed to post, comment and report	NFR-6.Test T1 to T9,T15.

8. AI usage log

Date	Tool / model	What we asked	What AI produced	What we changed / verified	What we learned

29 August 2026	Claude	Requested help structuring and drafting FRs/NFRs for a community website for people who are homesick, based on the StudyMate example structure	Draft of 5 FRs, 5 NFRs, use cases, and traceability matrix	(to be filled in based on actual edits: team name, the specific metrics actually tested for each NFR, the reporting threshold decided by the team for FR-5)	(to be filled in after real testing, describing what was learned)
19 October 2026	Claude	Asked Claude to help design and develop a daily motivational message feature for students living far from home.	Suggested a function that displays different motivational messages each day to encourage students and help them feel less lonely.	Reviewed and adjusted the messages to make them suitable for university students. Checked that the messages change daily and display correctly in the app.	Learned how AI can help develop app features and provide emotional support through daily motivational messages.

27 October 2026	claude	help split log- in to files, write log-in and sign-up code, connect it to the prototype, make a test plan, and draft the Final report and if SRS update.	The auth-*.js files, the connection code in the prototype, a test list, in draft text of the report and SRS v1.1.	Reviewed and tested the login and sign-up functions. Checked that users can register, log in, and connect to the database correctly. Fixed errors found during testing.	Learned how to use AI to organize code, develop authentication functions, connect the database, and test the application.

9. Self-check

- ☑ Every FR follows a clear trigger — system response format, tied to a real pain.
- ☑ Every FR has acceptance criteria you could actually test — 2 AC per FR.
- ☑ No FR is vague — 3. all FRs use clear and direct language.
- ☑ NFRs exist, including security / privacy. — Pass (Anonymous posting privacy and API response verification are fully defined in NFR-5.)
- ☑ Every FR traces back to a real problem — 7. show that every FR link to the real problem
- ☑ Scope and out-of-scope are stated — 2.1. and 6. Define the scope of works clearly.
- ☑ Use-case list matches the FR list — 5. has 1 use case per 1 FR.
- ☑ AI log — 8. clearly shows AI usage and how it was fixed.

10. Team and roles

Member	Student ID	Part
1.Thanaporn Dandon	6931503041	1.Log-in 2.SRS 3.Report Screen 4.Draft v.3_1 Create custom tags Report post system and Multi-tag selection&search 5. Merge Draft v.7
2.Pathanin Tamma	6931503110	1. SRS 2. Post & edit post 3. Comment on post 4. Filter posts by tag 5. Anonymous post
3. Ranchiya Jiwarangprapawong	6931503115	1. SRS 2. Sign-up
4. Sophitsuda Phonphanbuakhli	6931503120	1. Daft 3.3 added multi-level comment replies & edit comments 2. Log-out 3. SRS

5. Phontipa Chomnanon	6931503113	1. SRS 2. Draft 3.4 edit+delete post 3. Draft 5 Separate history for public&anonymous 4. Draft 6 Daily question reference in post creation